Results in Materials 17 (2023) 100363

Contents lists available at ScienceDirect

Results in Materials

journal homepage: www.sciencedirect.com/journal/results-in-materials

Design of asymmetric-structured metasurfaces for smart windows

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ARTICLE INFO

Keywords:
Metasurface
Smart windows
Micro-nano array structures
Multilayer
Fully-connected network

ABSTRACT

Smart windows with high visible transmission can control the absorption or transmission spectrum of solar irradiation based on seasonal variations and personal preferences, allowing buildings to save energy. However, it is a considerable challenge to design a metasurface with high solar modulation ability (ΔA_{sol}), integral luminous transmittance (T_{lum}), transmission light's color rendering index (Ra) and simple structure for smart windows. In this work, we design and optimize the asymmetric-structured multilayer and micro-nano array by applying the method of scanning parameters and fully-connected network methods. The optimization results show that ΔA_{sol} and T_{lum} negatively correlate when the transmitted light of these two structures has a high Ra. This conclusion of designing high-performance asymmetric-structured metasurfaces provides essential guidance for designing future smart windows.

1. Introduction

In the last decades, global warming and the energy crisis have attracted extensive interest because of the massive combustion of fossil fuels. Building energy consumption is currently responsible for 40–50% of total world energy consumption for light, ventilation, heating, and air conditioning [1,2]. Windows are the least energy-efficient structure because much heat and light are gained or lost via windows. Metasurfaces for smart windows artificially vary the radiation direction of electromagnetic waves entering or exiting buildings depending on seasonal variations and user preferences, achieving passive radiative cooling [3–5] or heating [6,7] of the buildings, alleviating energy consumption of buildings. Passive radiative cooling is environment-friendly technology that can successfully reflect the solar irradiance (λ = 0.3–2.5 μ m) with a broad "atmospheric window" (λ = 8–13 μ m) emission, making it suitable for implementation in buildings [8], automobiles [9], and human cooling [10,11]. Because of complex preparation technology, high cost, and most importantly, the low radiative cooling power (ideal radiative cooling power = 130 W/m^2 at 300 K [12]), the practical application of radiative cooling is seriously hindered [13]. The strength of solar energy with 1000 W/m^2 [14] leaves enough space for human beings to manipulate it. Many smart windows operating in the solar spectrum can selectively modify the transmittance, absorption, or reflection of sunlight (mainly in the visible and near-infrared ranges) using electrochromic [15], mechanically induced opacity [16,17], photochromic [18], thermochromic [19], driven by triboelectric nanogenerator (TENG) [20–22], and asymmetric structure [23–27]. However, most smart windows based on electrochromism and mechanically-induced opacity usually require auxiliary equipment and power consumption, and the photochromic smart windows cannot respond to extreme weather such as cloudy and foggy, while the thermochromic smart windows are heavily reliant on ambient temperature.

By contrast, smart windows relying on the asymmetric structure by rotating devices can successfully control the reflection, absorption or transmission of the solar spectrum without extra power supply and are independent of the external environment, giving them a considerable advantage. The asymmetric structure has different impedance for front and back side illumination, realizing directional optical absorption, reflection, or transmission. For instance, Buton et al. [25] exhibited asymmetric nanostructured metallic membranes with a 9-fold front-to-back absorption contrast. Jönsson et al. [26] presented an asymmetric structure made up of arrays of multielement gold antennas on glass and investigated the effect of the number of nanoparticles on directional optical absorption. Furthermore, Goodarzi et al. [28] took

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https://doi.org/10.1016/j.rinma.2023.100363
Received 4 August 2022; Received in revised form 8 December 2022; Accepted 5 January 2023
Available online 25 January 2023
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the court on the roles of asymmetry and interaction of nanoparticles in directional optical responses. Recently, there have been many reports on grating [27,29,30], moth-eye [31] and pyramidal [32–34] asymmetric nanostructures. However, the grating structure is related to the polarization state. It is challenging to prepare the moth-eye and pyramidal nanostructures. These issues will hinder the development and application of these structures. In recent research, we designed a low-cost copper plasmonic transparent metasurface with a maximum 25% absorption difference from different sides [23]. However, this nanostructure has a low Ra and ΔA_{sol}, which leads to the design can not being applied to smart windows. In addition, we presented an asymmetric-structured transparent metasurface with a high average visible transmission and extensive solar modulation [24]. However, this metasurface has a complicated structure, while the design process is not described in detail. Therefore, a detailed scheme for designing metasurface-based smart windows should be clarified urgently, and metasurfaces with large ΔA_{sol}, high T_{lum} and distinguished Ra are highly desirable.

Here, multilayer structures with excellent performance for smart windows are proposed, which do not depend on external energy and environmental conditions. Choose the combination of metals with high and low weighted solar absorption efficiency ($\overline{A_{sol}}$), which can successfully boost ΔA_{sol}, while optimizing the metal thickness and the size of the FP cavity of the structure can obtain a high T_{lum} and Ra. Due to the asymmetry of structure and metal, the Ti (3 nm)/SiO_2 (120 nm)/Ag (12 nm) multilayer structure demonstrates excellent performance (ΔA_{sol} = 0.223, T_{lum} = 0.672, Ra = 91). To further boost the performance, the micro-nano array structure enhances the T_{lum} by reducing the duty cycle of metals with the fully-connected network. We proposed the multilayer film structure and the micro-nano array structures outperform most reported asymmetric-structured metasurfaces. The ΔA_{sol} and T_{lum} of the multilayer and micro-nano array structures have obvious constraints - ΔA_{sol} and T_{lum} negatively correlate when the transmitted light has a high Ra. This conclusion of designing high-performance visible transparent metasurfaces provides an important guiding significance for the future design of smart windows, enhancing the practical application ability of the device for icephobicity [35,36], counteracting fogging [37], heat preservation building, and so on.

2. Materials and methods

Fig. 1 illustrates the schematic of the application of visible transparent metasurface for smart windows. This smart window plays diverse functions under various conditions. In the hot summer, the transparent metasurface faces the inside of the building, and it blocks most of the near-infrared light to keep cool indoors while maintaining high T_{lum} to keep the room bright. In the chilly winter, the structure can be rotated 180° so that the transparent metasurface towards outward. Due to the asymmetry of the structure, the metal of the absorbing layer is set thinner, and the sunlight can easily go through the absorbing layer and form a Fabry cavity resonance in the two layers of metal (the specific analysis is in the part of the "Design for high ΔA_{sol}"). In this state, the metasurface reduces sunlight reflection and increases absorption, which can be used in icephobicity and counteracting fogging. So, the smart window can minimize the use of air conditioning and reduce power consumption, realizing the alleviation of the energy crisis.

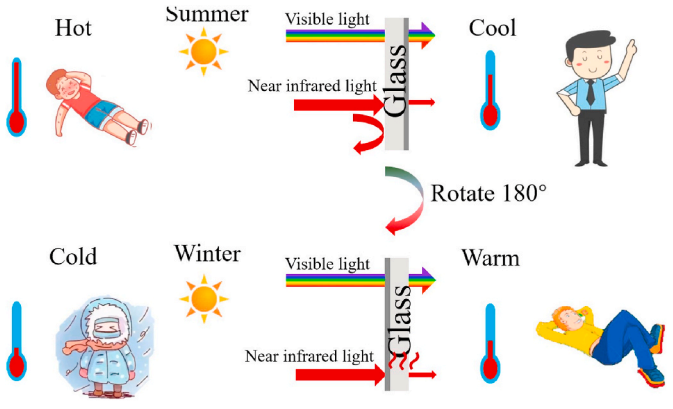


Fig. 1. The schematic of the transparent metasurface glazing applied in architectural windows.

For the multilayer thin-film structures, the transfer matrix method (TMM) [38] calculates the reflection, transmission and absorption. The interaction between each layer of film and electromagnetic wave can be determined by its characteristic matrix, and the characteristic matrix of *j* layer film can be described as:

$$M_j = \begin{bmatrix} \cos\delta_j & i\sin\delta_j/\eta_j \\ i\eta_j\sin\delta_j & \cos\delta_j \end{bmatrix} \tag{1}$$

where $\delta_j = 2\pi n_j d_j \cos\theta_j/\lambda$, $\eta_j = n_j\sqrt{\epsilon_0/\mu_0}\cos\theta_j$ (TE-polarization), and $\eta_j = n_j\sqrt{\epsilon_0/\mu_0}/\cos\theta_j$ (TM-polarization). λ is the wavelength, d_j, n_j and θ_j are the thickness, refractive index and propagation angle of the *j* layer medium, respectively. ϵ_0 is the vacuum permittivity, μ_0 is the vacuum magnetism. The characteristic matrix of *N*-layer structure is:

$$M = \prod_{j=1}^{N} M_j = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \tag{2}$$

According to Eq. (2), the reflection and transmission coefficients of the structure are:

$$r = \frac{\eta_0 m_{11} + \eta_0\eta_t m_{12} - m_{21} - \eta_t m_{22}}{\eta_0 m_{11} + \eta_0\eta_t m_{12} + m_{21} + \eta_t m_{22}} \tag{3}$$

$$t = \frac{2\eta_0}{\eta_0 m_{11} + \eta_0\eta_t m_{12} + m_{21} + \eta_t m_{22}} \tag{4}$$

where η_0 and η_t are the modified admittance of the input port and the outgoing port of electromagnetic wave, respectively. Therefore, the reflection and transmittance can be represented as $R = |r|^2$ and $T = |t|^2$, respectively. The absorption can be calculated as $A = 1 - R - T$.

For the micro-nano array structures, the finite-difference time-domain (FDTD) method is applied to electromagnetic calculations. The vertically TM-polarized wave is incident along the negative z-direction, and the polarization is along the x-direction. The periodic boundary conditions are used in the x and y-directions. A perfect matching layer (PML) is applied at the upper and bottom boundaries of the model.

The refractive index of metals (Fe, Gr, W, Ti, Au, Cu, Al, Ag) and SiO_2 are taken from experimental data [39]. ΔA_{sol} and T_{lum} are important performance metrics for smart windows and can be expressed by the formula:

$$A_{front-back} = \frac{\int \Phi_{sol}(\lambda)A_{front}(\lambda)d\lambda}{\int \Phi_{sol}(\lambda)d\lambda} - \frac{\int \Phi_{sol}(\lambda)A_{back}(\lambda)d\lambda}{\int \Phi_{sol}(\lambda)d\lambda} \tag{5}$$

$$\Delta A_{sol} = \left|A_{front-back}\right| \tag{6}$$

$$T_{lum} = \frac{\int \Phi_{lum}(\lambda)T(\lambda)d\lambda}{\int \Phi_{lum}(\lambda)d\lambda} \tag{7}$$

where the $A_{front}(\lambda)$ and $A_{back}(\lambda)$ are the absorption spectrum of structure when light is incident on the front and back sides, respectively. $\Phi_{sol}(\lambda)$ is the solar irradiance spectrum for an air mass of 1.5 [40], $\Phi_{lum}(\lambda)$ is the visibility function of the human eyes [41], which the formula can be expressed as:

$$\Phi_{lum}(\lambda) = 1.019 * e^{-0.0002854*(\lambda-559)^2} \tag{8}$$

The weighted solar absorption efficiency ($\overline{A_{sol}}$) can be calculated by:

$$\overline{A_{sol}} = \int_{300}^{4000} \Phi_{sol}(\lambda)A(\lambda)d\lambda \tag{9}$$

3. Design process and discussion

3.1. Design for high ΔA_{sol}

Undoubtedly, the higher the ΔA_{sol} of the smart windows, the wider its application. In this study, we start with the simplest structure. A simple asymmetric-structured metasurface composed of one layer of metal located on the SiO_2 (500 nm) substrate is shown in Fig. 2(a). Previous studies have reported a similar structure [25], and the ΔA_{sol} is caused by the asymmetry of the metal/dielectric structure. Table 1 displays the ΔA_{sol} and T_{lum} of various metals (10 nm) plated on SiO_2 (500 nm) substrate to examine the influence of different metal materials on ΔA_{sol}. The Cr/SiO_2 structure has the highest ΔA_{sol} (0.151) among these structures. Fig. 3(a) shows the R_{front}, R_{back}, T_{front}, T_{back}, A_{front} and A_{back} of the Cr/SiO_2 structure. Compared with R_{front}, R_{back} possesses a lower value because of the antireflection effect of SiO_2. Due to time reversal symmetry [25], the transmittance illuminated from the front side (T_{front}) is equal to that beamed from the back side (T_{back}), and the both front and back transmission curves overlap. The maximum ΔA_{sol} is only 0.151 when the thickness of Cr (h_{Cr}) changes in the range of 0–100 nm, as shown in Fig. 3(b). Therefore, this simple structure with a low ΔA_{sol} in Fig. 2(a) can not meet the practical application requirements.

To increase ΔA_{sol}, we replaced the metal/insulator nanocomposite (Fig. 2(a)) with the classical MIM structure (Fig. 2(b)), and the structure is coated with a 50 nm PMMA film, which can avoid metal oxidation and reduce reflection. PMMA with low refractive index, low price, easy-to-fabricate, high stability, etc., is extensively used in optical design [42–44]. In previous studies, the relationship between T_{lum} and the application of different metal materials is not clearly explained. Therefore, the theoretical analysis and simulation are combined to expound on their mutual influence. Table 2 calculates the $\overline{A_{sol}}$ of 8 sufficiently thick metals using Eq. (9) and ranks them according to their magnitude. Due to lower impedance in the solar spectrum range, W, Ti, Fe and Cr have a relatively high $\overline{A_{sol}}$, typically used in broadband solar absorption. Ag has a high impedance in the solar spectrum range resulting in little absorption of sunlight. Therefore, Ag is commonly used as a substrate for reflected light, especially for daytime radiation cooling structures.

The $A_{front\text{-}back}$ of different metal material combinations in MIM structure are given in Table 3. $A_{front\text{-}back} < 0$ means that the metal-II possesses stronger absorption in the solar spectrum than metal-I, causing the $\overline{A_{sol}}$ of back irradiation is higher than $\overline{A_{sol}}$ of front irradiation. When metal-I and metal-II are the same metal materials, the asymmetry of the upper and bottom dielectric layers of the MIM structure causes the ΔA_{sol} is small but not equal to 0. When the $\overline{A_{sol}}$ of metal-I and metal-II differ significantly (e.g., W–Ag, W–Al, and Fe–Ag) can result in a high ΔA_{sol}. The MIM structure with a large difference in $\overline{A_{sol}}$ causes larger ΔA_{sol}, which obtains wider application as smart windows.

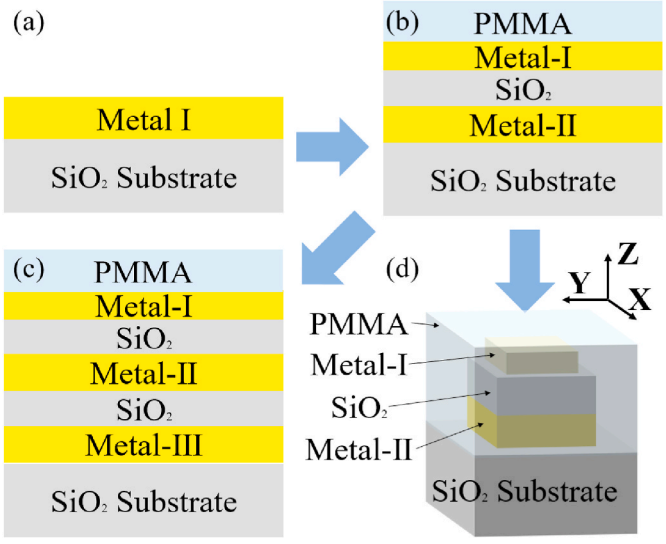


Fig. 2. Design process of transparent metasurface with high ΔA_{sol}, T_{lum} and Ra.

In addition, the thickness of the metal layer also significantly impacts ΔA_{sol}. Take the PMMA/Fe/SiO_2/Fe/substrate structure as an example (as shown in Fig. 4). If the $A_{front\text{-}back}$ is larger than 0, the metal-I and metal-II act as absorption and reflection layers, respectively. Then the thickness of metal-II can be fixed at 15 nm. When the metal-I is thin, PMMA/Fe/SiO_2/Fe/substrate can be regarded as PMMA/SiO_2/Fe/substrate, resulting in a small $A_{front\text{-}back}$. As the thickness of metal-I increases, the $A_{front\text{-}back}$ increases gradually due to the absorption of metal-I and the resonance of the FP cavity, and it reaches the peak when the thickness of metal-I is equal to 5 nm; When the thickness of metal-I increases further, and the thickness is close to metal-II's thickness, it is evident that the asymmetry caused by the thickness of the MIM structure is weakened, leading to a decrease in $A_{front\text{-}back}$; When metal-I is thicker than metal-II, metal-I and metal-II act as reflective and absorbing layers, respectively, resulting in $A_{front\text{-}back}$ less than 0. The thicknesses of the absorber and reflector layers in the MIM structure affect ΔA_{sol}, and ΔA_{sol} reaches the maximum when metal-I and metal-II differ by a few nanometers.

Through the above analysis, when the selected material and thickness of metal in the MIM structure, compared with the reflective layer metal, the absorption layer metal is preferable with higher $\overline{A_{sol}}$ and thinner thickness, which can achieve high ΔA_{sol}.

3.2. Design for high T_{lum}

It is a practical value that the T_{lum} of architectural windows is greater than 0.6. However, the T_{lum} was not considered when designing the high ΔA_{sol} above. The thickness of metal-I + metal-II is more than 20 nm (as shown in Fig. 4 (b)), and the $\overline{A_{sol}}$ is approximately zero. So, the thickness of metal has a significant influence on T_{lum}. In addition, the size of the FP cavity in the MIM structure also affects T_{lum}, and the formula expresses the FP cavity of MIM:

$$2\left(\frac{2\pi}{\lambda_{res}}\right)n_i t_i + \varphi_b + \varphi_t = 2\pi m \tag{10}$$

where λ_{res} is the resonance wavelength, n_i and t_i are the refractive index and thickness of the insulator layer, respectively. The m is an integer number that determines the order of cavity mode. The φ_b and φ_t are the phase shift acquired from the reflection from the bottom and top metal layers, respectively.

By changing the SiO_2 thickness, the position of the resonance absorption peak λ_{res} can be shifted to the near-infrared region, indirectly increasing the transmission in the 500–600 nm range, realizing a high T_{lum}. Taking the PMMA/Ag (10 nm)/SiO_2/Fe (5 nm)/substrate structure as an example, Fig. 5 gives the effect of SiO_2 thickness variation on A_{front}, A_{back} and T_{front}. The Ag and Fe are the reflection layer and absorption layer, respectively. Compared with the back-side incident (Fig. 5(b)), the thicker Ag obstructs the transmission of electromagnetic waves with the front side incident (Fig. 5(a)), resulting in significantly more minor changes in the FP cavity than the back side radiation. Fig. 5(c) shows the transmission spectrum (T_{front}) with the front incident light, and the time reversal symmetry can explain the T_{front} is parallel to the T_{back}. The change of SiO_2 thickness will also affect the visible range, directly involving T_{lum}. Fig. 5(d) reveals the relationship between ΔA_{sol} and SiO_2 thickness. When the thickness of SiO_2 is larger than 200 nm, the ΔA_{sol} has a small variation range due to the MIM with a high order of cavity mode, which results in the transmission spectrum with multiple peaks and valleys in the visible range. In conclusion, to design a transparent

Table 1
The ΔA_{sol} and T_{lum} of different metals (10 nm) are located on substrate (500 nm SiO_2).

Metal	W	Ti	Fe	Cr	Au	Cu	Al	Ag
ΔA_{sol}	0.142	0.134	0.141	0.151	0.034	0.031	0.063	0.009
T_{lum}	0.277	0.381	0.335	0.202	0.679	0.626	0.119	0.61

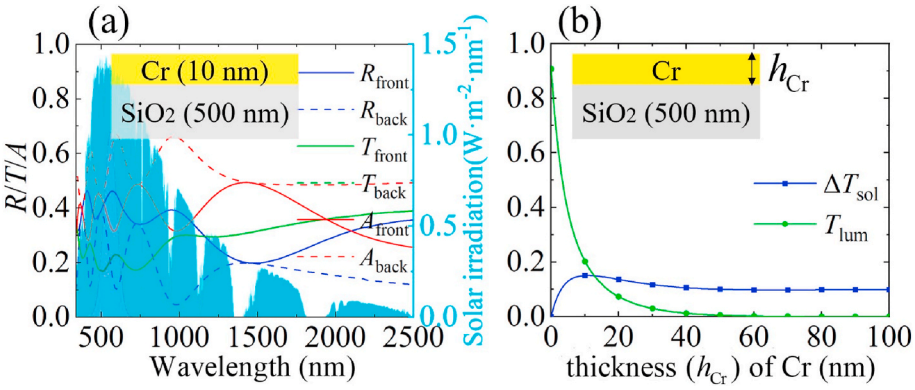


Fig. 3. (a) The reflection (R_{front}, R_{back}), transmission (T_{front}, T_{back}) and absorption (A_{front}, A_{back}) of Cr (10 nm)/SiO_2 structure, which was illuminated from the front and back side, respectively. The filled areas in light cyan and light purple represent the solar radiation spectrum (AM is 1.5) and the visibility function of human eyes, respectively. Due to time reversal symmetry [25], both T_{front} and T_{back} curves overlap. (b) ΔA_{sol} and T_{lum} of Cr/SiO_2 structure as a function of Cr thickness (h_{Cr}). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 2
The $\overline{A_{sol}}$ of various metals with sufficient thickness.

Metal	W	Ti	Fe	Cr	Au	Cu	Al	Ag
$\overline{A_{sol}}$	0.45	0.38	0.39	0.34	0.15	0.13	0.08	0.02

metasurface with high T_{lum}, it is necessary to consider the material and thickness of metals and the size of the FP cavity.

3.3. Design for high Ra

When the Ra of the transmission of the smart windows is larger than 85, it can meet people's demand to distinguish the original color of materials indoors through windows. In Fig. 6(a), the 1500 K, 3000 K and 10000 K black bodies have high Ra. However, the colors they showed can not be conducive to practical application. The equal-energy color is standard white, and the CIE chromaticity coordinate is (0.333, 0.333). The more the CIE chromaticity coordinate of the transmission spectrum close to (0.333, 0.333), the more metasurface owns a high Ra and white color. And this results in the metasurface can be more practical for smart windows.

Fig. 6(a) shows CIE chromaticity coordinate points (P_i) of 490 Ag/SiO_2/Fe structures' transmission with various thicknesses. The Ag and Fe thickness variation range is 3–21 nm, and the step size is 3 nm. The SiO_2 thickness variation range is 20–200 nm, and the step size is 20 nm. The d is the distance between Ag/SiO_2/Fe structure's transmission and equal-energy light (0.333, 0.333) on the CIE chromaticity coordinate. The red, green and black color points are represented $d \leq 0.02$, $0.02 < d \leq 0.04$, and $0.04 < d$, respectively. In Fig. 6(b), the P_M point of Ag (9 nm)/SiO_2 (120 nm)/Fe (3 nm)/substrate with $\Delta A_{sol} = 0.172$, $T_{lum} = 0.703$ and Ra = 93 of transmission light is undoubtedly the ideal structure for smart windows. It is preferred that the structure with the red point on the CIE chromaticity coordinate, attaching high Ra and white color of transmission sunlight.

3.4. Collaborative design of ΔA_{sol}, T_{lum} and Ra

Fig. 7 shows the metal material combinations (the scanning range of thickness is consistent with Fig. 6) of Table 3 with ΔA_{sol}, T_{lum} and Ra. Due to high $\overline{A_{sol}}$, the W, Ti, Fe and Cr metal combinations in Area a of Fig. 7 results in a low ΔA_{sol}. Therefore, these combinations are challenging to apply in practice. Due to the high and low $\overline{A_{sol}}$ metal material combinations in Area b and c of Fig. 7, these structures generally possess

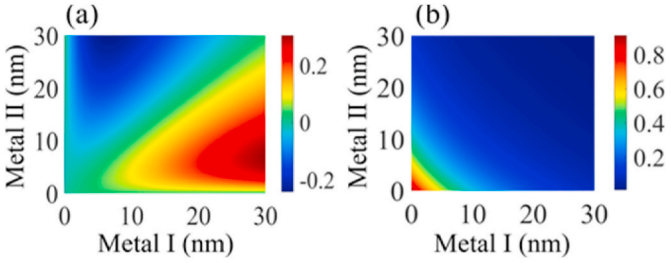


Fig. 4. The $A_{front\text{-}back}$ (a) and T_{lum} (b) versus the thickness of Metal-I and Metal-II of PMMA/Fe/SiO_2/Fe/substrate structure.

Table 3
The $A_{front\text{-}back}$ of various Metal-I (10 nm)/SiO_2 (100 nm)/Metal-II (10 nm)/substrate.

Metal-I	Metal-II							
	W	Ti	Fe	Cr	Au	Cu	Al	Ag
W	0.001	−0.057	−0.023	0.023	0.275	0.284	0.475	0.359
Ti	0.094	0.034	0.074	0.120	0.353	0.368	0.568	0.412
Fe	0.054	−0.010	0.030	0.075	0.327	0.342	0.533	0.393
Cr	−0.001	−0.063	−0.024	0.021	0.317	0.328	0.516	0.393
Au	−0.226	−0.284	−0.261	−0.265	0.019	0.032	0.020	0.032
Cu	−0.233	−0.296	−0.273	−0.271	0.007	0.020	0.025	0.020
Al	−0.440	−0.494	−0.466	−0.456	0.023	0.023	0.035	0.110
Ag	−0.302	−0.337	−0.319	−0.335	0.003	0.022	−0.075	0.007

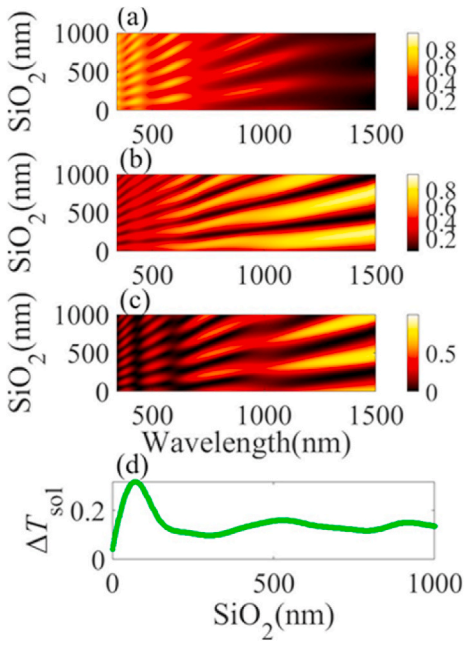


Fig. 5. The A_{front} (a), A_{back} (b), T_{front} (c) and ΔA_{sol} (d) versus the thickness of SiO_2 of Ag (10 nm)/SiO_2/Fe (5 nm) structure.

high ΔA_{sol}. The Au, Cu, Al and Ag with high impedance lead to a low ΔA_{sol} of metal material combinations in Area d of Fig. 7. A significant $\overline{A_{sol}}$ difference between metal-I and metal-II allows structure with excellent performance. And then, the asymmetric-structured metasurfaces can realize high ΔA_{sol}, T_{lum} and Ra by screening the appropriate metal and SiO_2 thickness. It is obvious that the Ag (12 nm)/SiO_2(120 nm)/Ti(3 nm) structure with excellent performance ($\Delta A_{sol} = 0.223$, $T_{lum} = 0.672$, Ra = 89). Fig. 8(a) and (b) show the structural diagram, the R, T and A of the Ag/SiO_2/Ti structure. The inset in Fig. 8(b) shows the CIE chromaticity coordinate of the structure's sunlight transmission. In addition, the multilayer film structure we designed does not depend on external energy and environmental conditions and only needs to rotate the window to realize the regulation of sunlight. The materials used are cheap, the preparation is relatively simple, and the substrate can be common glass, which is especially suitable for application in household and office windows. After mass production in the future, the cost of smart windows can be further reduced, and the application field can be further expanded.

3.5. Methods to improve performance

It can improve the performance of smart windows by changing the structure of Fig. 2(b). One way is adding the number of adjustable parameters by increasing the FP cavity numbers, as shown in Fig. 2(c). However, too many metal layers result in substantial structure absorption, causing a low $\overline{A_{sol}}$. What's more, multilayer metals are difficult to control the asymmetry of the structure, so this method is not feasible.

The other way is reducing the duty cycle of metal, as shown in Fig. 2 (d), and it can increase the transmission by 500–600 nm. The duty cycle of the multilayer thin-film structure in Fig. 2(b) is 100%, and it can reduce the duty cycle of one layer of metal, similar to our previous design [23]. It is more manipulation space when changing the duty cycle of two metal layers, just like our recent work [24]. It should be pointed out that the 50 nm PMMA layer is plated on the MIM structure, and its thickness is no longer changed during parameter optimization. The length and width of the metals are equal to avoid the effect of incident light polarization. However, the simulation time of this micro-nano array structure is larger than the multilayer structure. Therefore, a fully-connected network optimizes the structure to perform better.

3.5.1. Network prediction procedure

The micro-nano array structures have more optimization parameters and longer simulation time than multilayer structures. Hence, this work proposes a fully-connected network based on position coding for optimized micro-nano array structures, which saves simulation time by predicting data based on the truth values. Recently, data input mainly converts the binary value into the network [45]. In contrast, positional encoding expands input coordinates into a higher dimensional space, enabling the fully-connected deep network to learn faster. The positional encoding equation used there is:

$$\mathrm{Input}(p) = \left(\sin\left(2^0\pi p\right), \cos\left(2^0\pi p\right), \cdots, \sin\left(2^{L-1}\pi p\right), \cos\left(2^{L-1}\pi p\right)\right) \tag{11}$$

where the Input(·) represents the input parameter (which is normalized to lie in [0, 1]). The right side of the equation is the transformed high-frequency function, and L is set to 10.

As is shown in Fig. 9, the network inputs are h_1 (metal-I thickness), h_2 (SiO_2 thickness), h_3 (metal-II thickness), w_1 (metal-I width) and w_2 (metal-II width), and the network outputs are the ΔA_{sol}, T_{lum} and Ra.

During the network prediction process, it takes the micro-nano array structure Fe/SiO_2/Ag/substrate as an example. The variations and limits of the state are defined as follows: h_1 and h_3 = (20, 20, 60), h_2 = (40, 20, 80), w_1 and w_2 = (160, 20, 260). The three data in brackets represent the variables' lower limit, step size and upper limit. The total number of states is $3 \times 3 \times 3 \times 6 \times 6 = 972$, and the FDTD simulates the ΔA_{sol}, T_{lum} and Ra of these states. The 972 data are split into 80% for training, 10% for validation, and 10% for testing. The 972 states with different ΔA_{sol},

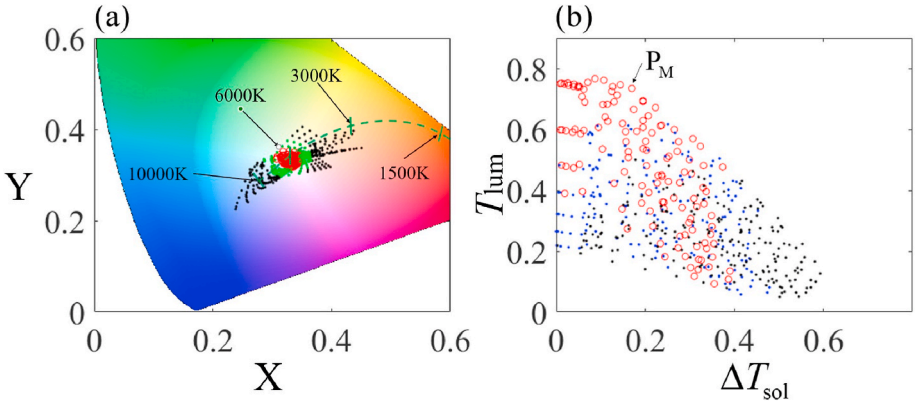


Fig. 6. (a) The CIE chromaticity coordinate of 810 states with various Ag/SiO_2/Fe structures. The red curve represents the black body at different temperatures. (b) The ΔA_{sol}, T_{lum} of 810 states. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

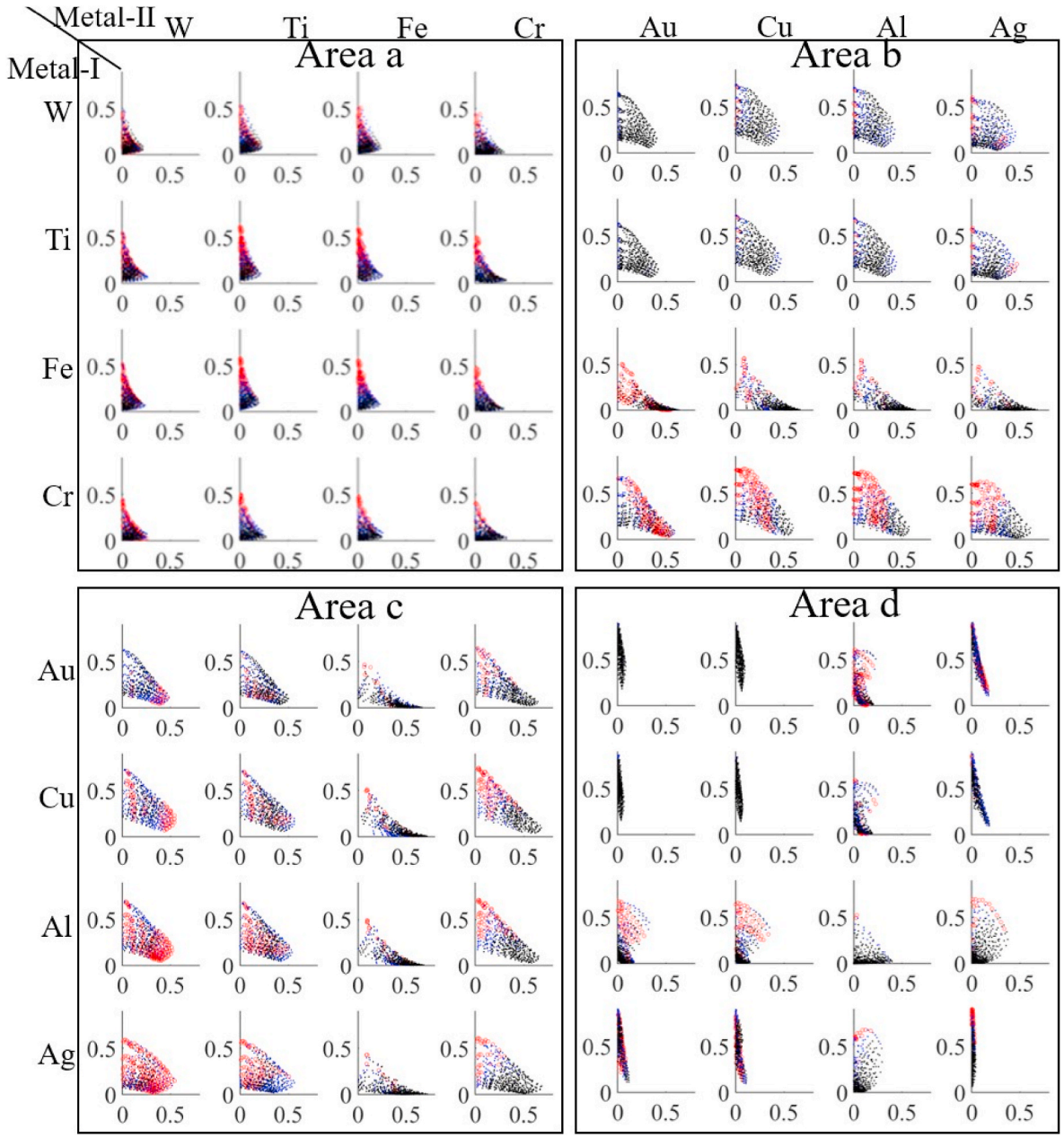


Fig. 7. The ΔA_{sol}, T_{lum} and Ra of MIM structure with different materials and structural dimensions. It can be divided into Area a, b, c and d according to the different combinations of larger $\overline{A_{sol}}$ metals (W, Ti, Fe, Cr) and smaller $\overline{A_{sol}}$ metals (Au, Cu, Al, Ag).

T_{lum} and Ra are shown in Fig. 10(a). Then the input data after training is 15,125, which is the combinations of various parameters of h_1, h_2, h_3, w_1 and w_2 (the parameter variation range is not changing, and the step size is 10 nm). The ΔA_{sol}, T_{lum} and Ra of 15,125 states are shown in Fig. 10 (b).

3.5.2. Network evaluation

Mean square error (MSE) is adopted to average the losses over the output, which is the sum of squared distances between truth values and predicted values. The MSE of ΔA_{sol}, T_{lum} and Ra are 0.000146, 0.000131, and 1.36, indicating the model's accuracy. The network prediction process is trained on an Nvidia GTX 1650 with 4 GB memory. The fully-connected network is set up with 150,000 iterations, with a training time of 3.07 min and above 1 ms to predict a data set. However, it takes 16 min to simulate a data set with FDTD. The results show that the network prediction increases the states and greatly reduces the simulation time while ensuring correct computation. For example, the ΔA_{sol}, T_{lum} and Ra of the micro-nano array structure Fe/SiO_2/Ag/substrate ($h_1 = 40$ nm, $h_2 = 40$ nm, $h_3 = 20$ nm, $w_1 = 160$ nm and $w_2 = 190$ nm) structure predicted by the network is in good agreement with the FDTD calculation, as shown in Table 4. Similarly, the network prediction method is also used to predict other micro-nano array structures, which is not given here for space reasons.

4. Comparison work

Fig. 11 shows the comparison of this work with previous smart windows based on asymmetric structure. The development trend of visible transparent metasurfaces for smart windows can be drawn from these earlier reports (black star dots) [23–27], and high ΔA_{sol}, T_{lum} and Ra are required for practical application. It is a meaningless pursuit of one-sided distinguished ΔA_{sol}, ultrahigh T_{lum} or high Ra. The red and

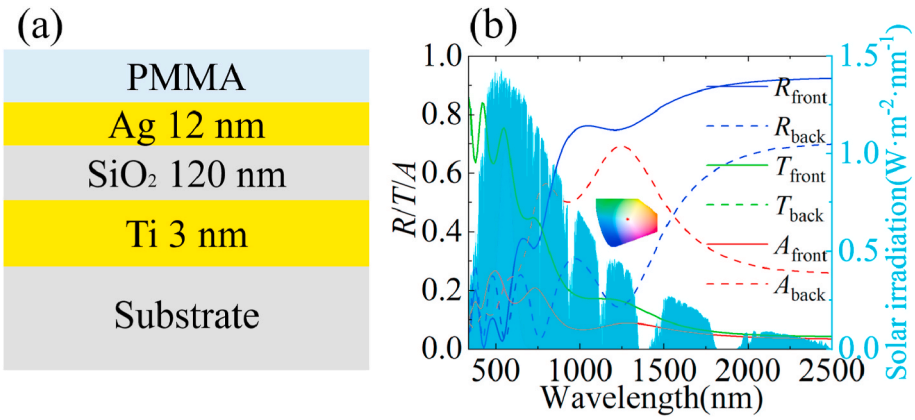


Fig. 8. The schematic (a), R_{front}, R_{back}, T_{front}, T_{back}, A_{front} and A_{back} (b) of the Al/SiO_2/Cu structure. The CIE chromaticity coordinate of transmission light is shown in the illustration. Due to time reversal symmetry [25], both T_{front} and T_{back} curves overlap.

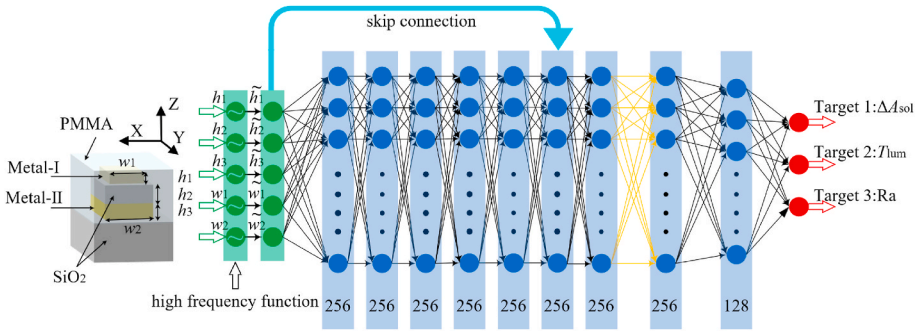


Fig. 9. A architecture of the fully-connected network. The input vectors, intermediate hidden layers and output vectors are shown in green, blue and red, respectively. The number inside each block represents the vector dimension. The black arrows indicate layers with rectified linear unit (ReLU) activations, and the orange arrows indicate layers with no ReLU activation. The Cyan arrow indicates the skip connection. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

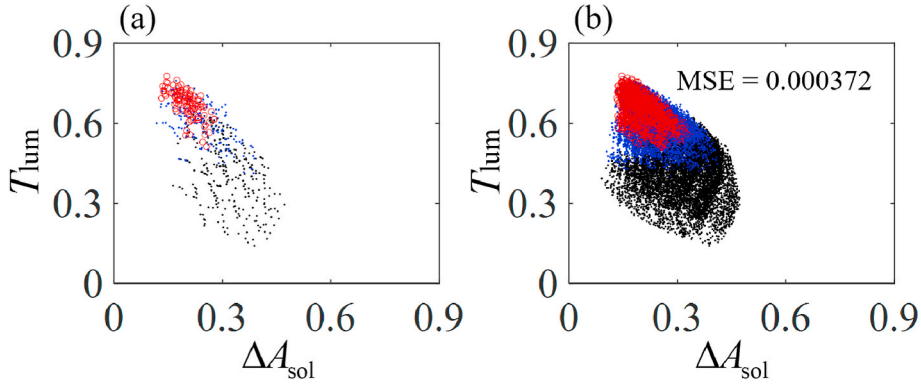


Fig. 10. (a) The ΔA_{sol}, T_{lum} and Ra for 972 states calculated by FDTD. (b) The network model predicted the ΔA_{sol}, T_{lum} and Ra for 15,125 states.

Table 4
Comparison of ΔA_{sol}, T_{lum} and Ra of the micro-nano array structure Fe/SiO_2/Ag/substrate calculated by FDTD and deep learning.

Method	ΔA	T_{lum}	Ra
FDTD calculation	0.189	0.752	87
Predicted data	0.192	0.753	89

blue dots in Fig. 11 indicate the multilayer film structures and micro-nano array structures with excellent performance after the above collaborative design of ΔA_{sol}, T_{lum} and Ra. The red and blue dotted lines indicate that the relationship between ΔA_{sol} and T_{lum} of the optimal multilayer and micro-nano array structures is negatively correlated. It is worth reminding that the Ra of these structures (the red and blue dots) is greater than 85, and the transmission color of these structures is very close to white ($d \leq 0.02$ in CIE chromaticity coordinate).

The previous work [23,25] has low ΔA_{sol} and T_{lum}, making them difficult to apply in energy-saving windows in artificial buildings. The structure of [26,27] possesses a relatively high T_{lum}. However, the low ΔA_{sol} results in those devices cannot apply in smart windows. The multilayer structures (red dots) outperform previously reported work in this work [23,25–27]. For example, the Ag (12 nm)/SiO_2 (120 nm)/Ti (3 nm) multilayer structure (P_A in Fig. 11) possesses excellent performance ($\Delta A_{sol} = 0.223$, $T_{lum} = 0.672$, and Ra = 89). This kind of metasurface with superior performance and simple structure is beneficial in smart windows. Compared with micro-nano array structures, the multilayer structures attach a wider modulation range in ΔA_{sol}. While the micro-nano array structures retain a higher T_{lum} in the 0.1–0.32 range of ΔA_{sol}. To improve the T_{lum}, we reduce the duty cycle of metal of the micro-nano array structures, and the performance is better than in

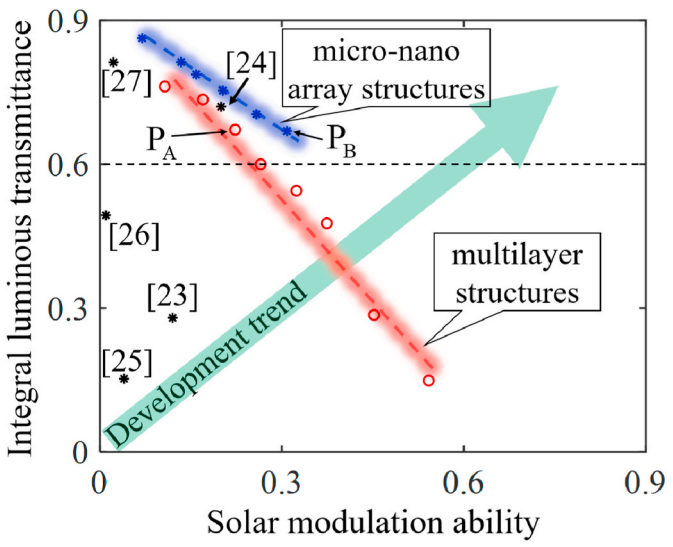


Fig. 11. Comparison of this work with recently reported smart windows based on asymmetric structures.

multilayer structures, including our previous work [24]. Compared with P_A (ΔA_{sol} = 0.223), the structure of P_B in Fig. 11 has a higher ΔA_{sol} (0.308). In practical applications, if the customer requires smart windows with a high ΔA_{sol} (>0.35), the multilayer structures is more suitable; If the glass requires a high value of T_{lum} along with a high ΔA_{sol} in the buildings, the micro-nano array structures are preferred.

These conclusions have a guiding role in designing transparent asymmetric-structured metasurfaces for smart windows in the future.

5. Conclusion

We proposed and optimized two types of transparent asymmetric-structured metasurfaces (the multilayer film and the micro-nano array structures) by applying the method of scanning parameters and the fully-connected network. There is a negative correlation between ΔA_{sol} and T_{lum} of the two types of structures when the Ra of transmitted light is larger than 85. This conclusion for designing transparent metasurfaces with high performance offers an important guiding significance for designing smart windows. And the performances of our proposed structures outperform most reported work based on asymmetric structure. Besides, the proposed easy-to-fabricate, high ΔA_{sol}, T_{lum}, and Ra smart windows have promising prospects in icephobicity, counteracting fogging, and other energy management applications.

Credit author statement

Xueyu Wang: Writing –review & editing, Validation, Supervision, Project administration. Shuo Chen: Writing –review & editing. Yumin Liu: Writing –review & editing, Validation, Supervision, Project administration, Funding acquisition. Tiesheng Wu: Writing –review & editing, Validation, Supervision, Project administration, Funding acquisition. Jing Li: Writing –review & editing, Supervision. Danfeng Zhu: Writing –review & editing, Supervision. Yu Wang: Writing –review & editing, Supervision. Yuhang Sun: Writing –review & editing, Supervision. Hongjie Fan: Writing –review & editing, Supervision. Han Ye: Writing –review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research was funded by the National Natural Science Foundation of China (NSFC) (61875021); Natural Science Foundation of Beijing (2192036); The Fundamental Research Funds for the Central Universities; Guangxi Key Laboratory of Wireless Wideband Communication and Signal Processing; BUPT Action Plan Project (ZDYY202102-1).

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